MEDIA

Chapter 20

Using the Inspector in the Media Page

The Inspector holds all the controls to modify, resize, retime, and generally adjust anything related to a clip, transition, or effect on the Media page Timeline.

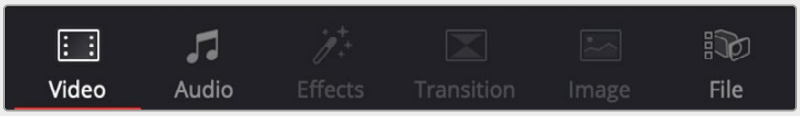
Contents

Using the Inspector

The Inspector has been redesigned to make it easier to find specific controls and to adjust common settings for your clips. Instead of a long vertical list, different aspects of the Inspector have now been organized into panels, with each controlling specific grouped sets of parameters for your clip.

The Inspector is activated by clicking on the Inspector Panel in the upper-right section of the User Interface toolbar. The Inspector is broken up into individual Video, Audio, Effects, Transition, Image, and File panels. Inspector panels that are not applicable to your clip or selection are grayed out.

The Inspector Panel icon in the upper right of the UI toolbar



The Inspector panels showing Video, Audio, and File parameters available for adjustment; others are grayed out.

Methods of using controls in the Inspector:

- **To activate or deactivate a control:** Click the toggle to the left of the control's name. The orange dot on the right means the control is activated. A gray dot on the left means the control is deactivated.
- **To reveal a control's parameters:** Double-click the control's name.
- **To reset controls to their defaults:** Click the reset button to the right of the control's name.

Adjusting Media Pool Clips in the Inspector

You can directly modify Media Pool clips in the Inspector, before you edit those clips into a timeline. This allows you to change the parameters of source media so that clips that are subsequently edited into a timeline carry those new settings with it. For example, you can prepare your material prior to editing by changing the clip's file and RAW settings, adjusting the audio levels and EQ, or assigning it a specific lens correction, etc. Once modified, any part of that clip would have the correct Inspector parameters already in place when you edited them into your timeline.

To adjust Media Pool clips in the Inspector:

1. Select one or more clips in the Media Pool Panel of either the Media, Cut, Edit, or Fairlight pages.
2. Open the Inspector panel, and adjust any parameters in the Video, Audio, Image and File tabs.

These parameter changes are stored with the Media Pool clip, and will be carried over when any part of that clip is edited into the Timeline. Of course, each clip's Inspector parameters can be further modified once it's in the Timeline, and those Timeline parameters are independent from the Media Pool Inspector settings. This means that any further adjustments you make to the clip in the Timeline do not affect that same clip's adjustments already in the Media Pool.

Video

The Video Panel of the Inspector exposes a vast array of controls designed to manipulate the size, speed, and opacity of your clips.

Transform

The Transform section of the Video Inspector panel

The Transform group includes the following parameters for resizing and repositioning your clips:

- **Zoom X and Y:** Allows you to blow the image up or shrink it down. The X and Y parameters can be linked to lock the aspect ratio of the image, or released to stretch or squeeze the image in one direction only.
- **Position X and Y:** Moves the image within the frame, allowing pan and scan adjustments to be made. X moves the image left or right, and Y moves the image up or down.
- **Rotation Angle:** Rotates the image around the anchor point.
- **Anchor Point X and Y:** Defines the coordinate on that clip about which all transforms are centered.
- **Pitch:** Rotates the image toward or away from the camera along an axis running through the center of the image, from left to right. Positive values push the top of the image away and bring the bottom of the image forward. Negative values bring the top of the image forward and push the bottom of the image away. Higher values stretch the image more extremely.
- **Yaw:** Rotates the image toward or away from the camera along an axis running through the center of the image from top to bottom. Positive values bring the left of the image forward and push the right of the image away. Negative values push the left of the image away and push the right of the image forward. Higher values stretch the image more extremely.
- **Flip Image:** Two buttons let you flip the image in different dimensions.

  Flip Horizontal control: Reverses the image along the X-axis, left to right.

  Flip Vertical control: Reverses the clip along the Y-axis, turning it upside down.

Cropping

The Cropping section of the Video Inspector panel

The Video Inspector controls the image's cropping parameters:

- **Crop Left, Right, Top, and Bottom:** Lets you cut off, in pixels, the four sides of the image. Cropping a clip creates transparency so that whatever is underneath shows through.
- **Softness:** Lets you blur the edges of a crop. Setting this to a negative value softens the edges inside of the crop box, while setting this to a positive value softens the edges outside of the crop box.
- **Retain Image Position:** Clicking this checkbox will lock the crop parameters in place when you resize the image using the Transform tool above. Unchecking this box will scale and position the crop as well as the image.

Dynamic Zoom

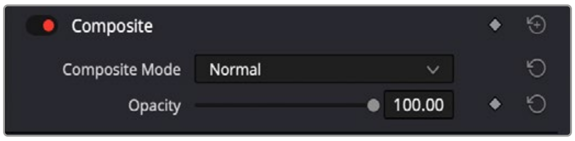
The Dynamic Zoom section of the Video Inspector panel

The Dynamic Zoom controls, which are off by default, make it fast and easy to do pan and scan effects to zoom into or out of a clip. Turning the Dynamic Zoom group on activates two controls in the Inspector that work hand-in-hand with the Dynamic Zoom onscreen adjustment controls.

> For more information on using the Dynamic Zoom controls, *see Chapter 50, "Compositing and Transforms in the Timeline."*

- **Dynamic Zoom Ease:** Lets you choose how the motion created by these controls accelerates. You can choose from Linear, Ease In, Ease Out, and Ease In and Out.
- **Swap:** This button reverses the start and end transforms that create the dynamic zoom effect.

Composite



The Composite section of the Video Inspector panel

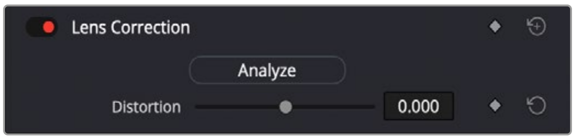
Composite modes can be used to combine clips that are superimposed over other clips in the Timeline.

- **Composite Mode:** This selects the type of composite mode to combine the superimposed clips. The default "Normal" means no compositing mode is applied.

> For more information on Composite Modes, *see Chapter 50, "Compositing and Transforms in the Timeline."*

- **Opacity:** This slider makes a clip more or less transparent in addition to compositing already being done.

Lens Correction

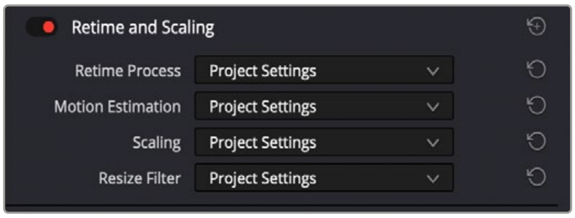


The Lens Correction section of the Video Inspector panel

The Lens Correction group (only available in Resolve Studio) has two controls that let you correct for lens distortion in the image, or add lens distortion of your own.

- **Analyze:** Automatically analyzes the frame in the Timeline at the position of the playhead for edges that are being distorted by wide angle lens. Clicking the Analyze button moves the Distortion slider to provide an automatic correction. If you're analyzing a particularly challenging clip, a progress bar will appear to let you know how long this will take.
- **Distortion:** Dragging this slider to the right lets you manually apply a warp to the image that lets you straighten the bent areas of the picture that can be caused by wide angle lenses.
If you clicked the Analyze button and the result was an overcorrection, then dragging this slider to the left lets you back off of the automatic adjustment until the image looks correct.

Retime and Scaling



The Retime and Scaling section of the Video Inspector panel

The Retime and Scaling group has four parameters that affect retiming quality and clip scale:

- **Retime Process:** Lets you choose a default method of processing clips in mixed frame rate timelines and those with speed effects (fast forward or slow motion) applied to them, on a clip-by-clip basis. The default setting is "Project Settings," so all speed-effected clips are treated the same way.

> There are three options: Nearest, Frame Blend, and Optical Flow, which are explained in more detail in the Speed Effect Processing section of Chapter 51, "Speed Effects."

- **Motion estimation mode:** When using Optical Flow to process speed change effects or clips with a different frame rate than that of the Timeline, the Motion Estimation pop-up lets you choose the best-looking rendering option for a particular clip. Each method has different artifacts, and the highest quality option isn't always the best choice for a particular clip. The default setting is "Project Settings," so all speed-effected clips are treated the same way. There are several options. The "Standard Faster" and "Standard Better" settings are the same options that have been available in previous versions of DaVinci Resolve. They're more processor efficient and yield good quality that are suitable for most situations. However, "Enhanced Faster" and "Enhanced Better" should yield superior results in nearly every case where the standard options exhibit artifacts, at the expense of being more computationally intensive, and thus slower on most systems.
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Better” should yield superior results in nearly every case where the standard options exhibit artifacts, at the expense of being more computationally intensive, and thus slower on most systems.

- “Speed Warp Faster and “Speed Warp Better” are available for even higher-quality slow motion effects using the DaVinci Neural Engine. Your results with this setting will vary according to the content of the clip, but in ideal circumstances this will yield higher visual quality with fewer artifacts than even the Enhanced Better setting.

- **Scaling:** Lets you choose how clips that don’t match the current project resolution are handled on a clip-by-clip basis. The default setting is “Project Settings,” so that all mismatched clips use the same method of being automatically resized.

  However, you can also choose an individual method of automatic scaling for any clip. The options are Crop, Fit, Fill, and Stretch; for more information see the 2D Transforms section *see Chapter 152, “Sizing and Image Stabilization.”*

- **Resize Filter:** For clips that are being resized in any way, this setting lets you choose the filter method used to interpolate image pixels when resizing clips. Different settings work better for different kinds of resizing. There are four options:

  Sharper: Usually provides the best quality in projects using clips that must be scaled up to fill a larger frame size, or scaled down to HD resolutions.

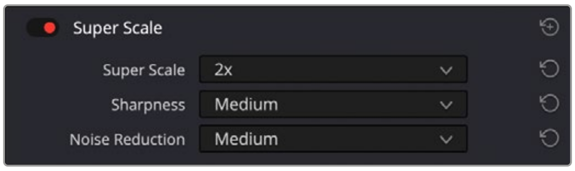
  Smoother: May provide higher quality for projects using clips that must be scaled down to fit an SD resolution frame size.

  Bicubic: While the Sharper and Smoother options are slightly higher quality, Bicubic is still an exceptionally good resizing filter and is less processor intensive than either of those options.

  Bilinear: A lower quality setting that is less processor intensive. Useful for previewing your work on a low-performance computer before rendering, when you can switch to one of the higher quality options.

  Other Resize Methods: A selection of specific resize algorithms is available if you need to match them to other VFX workflows.

Super Scale



The Super Scale parameters

For instances when you need higher-quality upscaling than the standard Resize Filters allow, you can now enable one of the “Super Scale” options in the Inspector. Unlike using one of the numerous scaling options in the Edit, Fusion, or Color pages, Super Scale actually increases the source resolution of the clip being processed, which means that clip will have more pixels than it did before and will be more processor-intensive to work with than before, unless you optimize the clip (which bakes in the Super Scale effect into the optimized media) or cache the clip in some way.

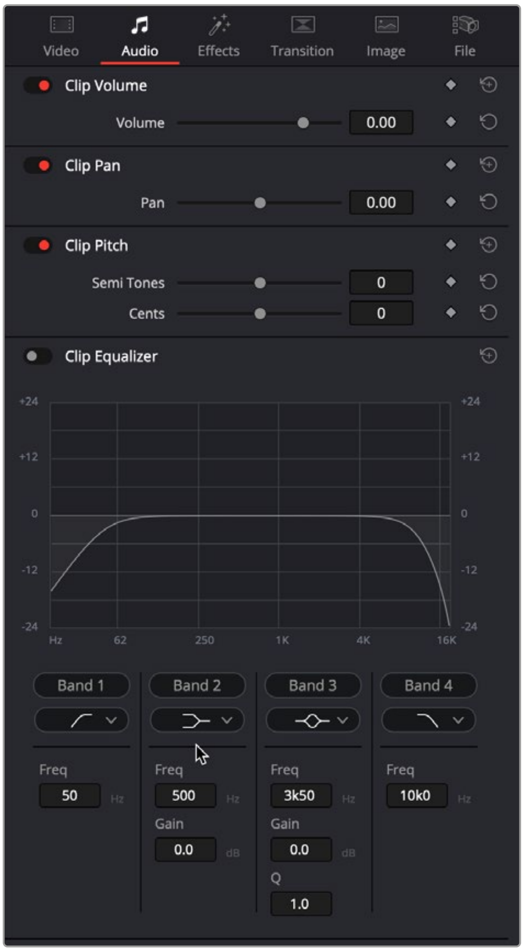
For more detailed information on Super Scale, *see Chapter 11, “Image Sizing and Resolution Independence.”*

Using Super Scale in the Inspector is functionally equivalent to setting the same controls for the media clip in the Clip Attributes. This means that changing this setting affects all of the additional edits referencing the selected media as well.

The Super Scale group has the following parameters that affect the quality and clip scale:

- **Super Scale:** Lets you choose the amount of scaling required. The options are 2-3-4x and 2-3-4x Enhanced.
- **Sharpness:** Lets you choose the amount of detail in the scaling. This is limited to Low, Medium, or High, unless the Super Scale mode is set to Enhanced, which allows you to apply variable sharpness. You will want to balance this setting against Noise Reduction.
- **Noise Reduction:** Lets you choose the amount of noise reduction in the scaling. This is limited to Low, Medium, or High, unless the Super Scale mode is set to Enhanced, which allows you to apply variable noise reduction. You will want to balance this setting against Sharpness.

Audio



The Audio Inspector parameters

The Audio tab contains four commonly used audio controls for video editing purposes, including Clip Volume, Clip Pan, Clip Pitch, and Clip Equalizer.

Clip Volume: Each clip has a single volume control that corresponds to the volume overlay over each audio clip.

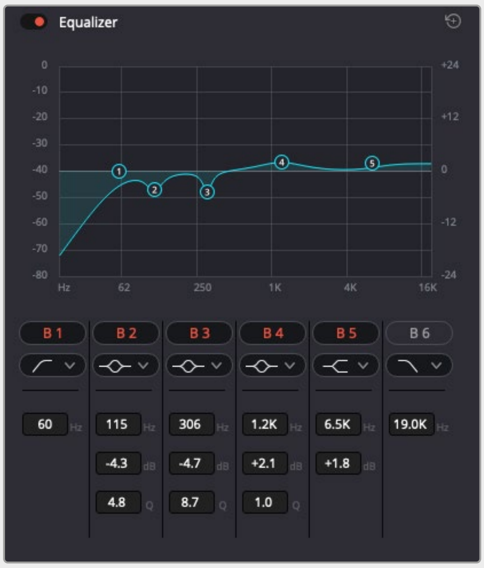
Clip Pan: (Only exposed for clips) A simple Pan slider that controls stereo panning.

Clip Pitch: Lets you alter the pitch of a clip without changing the speed. Two sliders let you adjust clip pitch in semi tones (large adjustments, a twelfth of an octave) and cents (fine adjustments, 100th of a semi tone).

Equalizer

This six-band parametric equalizer has both graphical and numeric controls for boosting or attenuating different ranges of frequencies within that clip, before it even gets to the EQ built into the Mixer. Each band has controls for the filter type (Bell, Lo-Shelf, Hi-Shelf, Notch), Frequency, Gain, and Q-factor (sharpness of the band), with the available controls for each band of EQ changing depending on the filter type.

TIP: Clip EQ settings can be copied and pasted from one Clip EQ to another, and between Clip EQ and Track EQ instances.



Clip EQ in the Audio Inspector

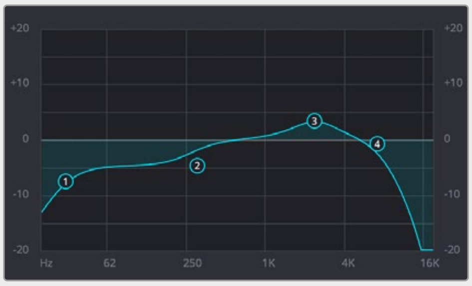
Master EQ Controls

The Equalizer window has the following overall controls located in the upper right and left corners.

- **Enable button:** Turns the overall EQ effect off and on, without resetting the controls.
- **Reset button:** Resets all controls of the EQ window to their defaults.

Clip EQ Graph

This graph displays the current EQ curve with handles that correspond to each enabled EQ band listed below. You can drag any numbered handle to boost or attenuate the range of frequencies governed by that band, using whatever type of equalization that band has been set to..



The Clip EQ graph with user-draggable handles

Dragging the numbered handles on this graph in turn modifies the parameters of the corresponding band, and changing each band's parameters will also alter the EQ graph, which serves the additional purpose of providing a graphical representation of the equalization being applied to that track.

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Bands 1 and 6

The outer two sets of band controls let you make high-pass and low-pass adjustments, if necessary.

- **Band enable button:** Turns each band of EQ on and off.
- **Band filter type:** Bands 1 and 6 can be switched among five specific filtering options for processing the lowest or highest frequencies in the signal.
- **For Band 1, these include (from top to bottom):** Lo-Shelf, Bell, Notch, Hi-Shelf, and Hi-Pass.
- **For Band 6, the options are (from top to bottom):** Lo-Pass, Lo-Shelf, Bell, Notch, and Hi-Shelf.
- **Freq:** Adjusts the center frequency of the EQ adjustment.
- **Gain:** Adjusts the amount by which the affected frequencies are impacted. Negative values attenuate those frequencies, while positive values boost those frequencies.
- **Q Factor:** This parameter adjusts the width of affected frequencies and appears whenever the Bell option is selected. Lower values include a wider range of frequencies; higher values include a narrower range of frequencies.

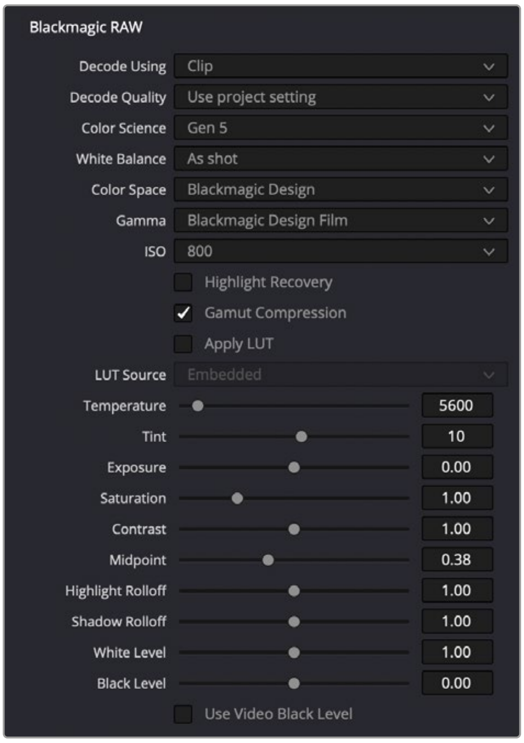
Bands 2 through 5

These band controls let you make a wide variety of equalization adjustments.

- **Band enable button:** Turns each band of EQ on and off.
- **Band filter type:** Offers four filtering options (from top to bottom): Lo-Shelf, Bell, Notch, and Hi-Shelf.
- **Frequency:** Adjusts the center frequency of the EQ adjustment.
- **Gain:** Adjusts the amount by which the affected frequencies are altered. Negative values attenuate those frequencies, while positive values boost those frequencies.
- **Q Factor:** This parameter appears when the Bell setting is selected. It adjusts the width of affected frequencies. Lower values include a wider range of frequencies; higher values narrow the range of frequencies.

NOTE: There are many more refined plugins and effects for audio clips in the Audio FX library. If you apply any of these, the controls will appear in the Inspector’s Effects tab Audio section, instead of here.

Image

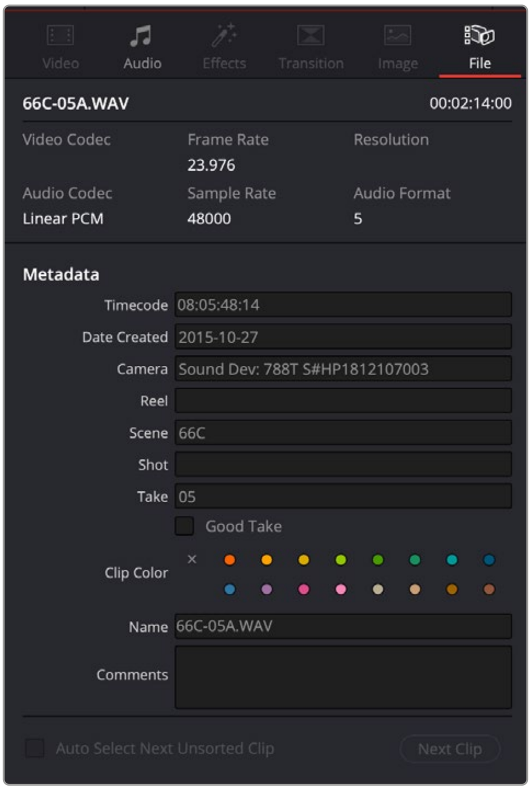


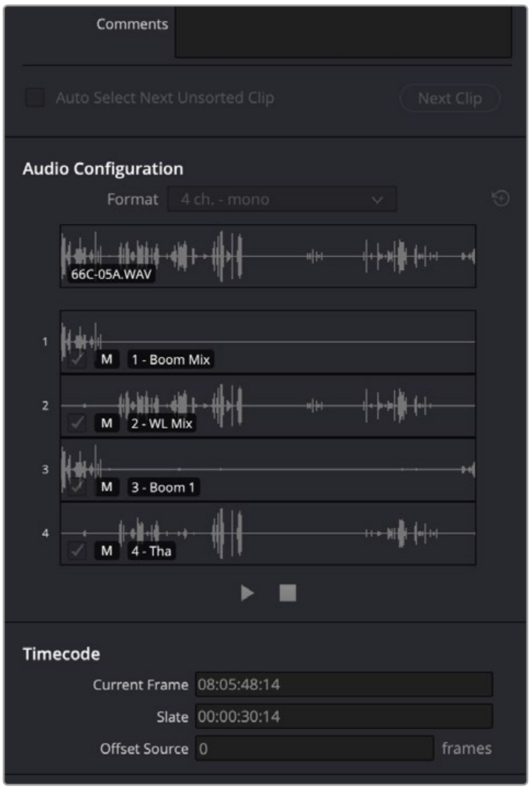
The Image Inspector controμls for BRAW footage

The Image panel contains groups of parameters that correspond to every camera raw media format that's supported by DaVinci Resolve. Using these parameters in the Image panel, you can override the original camera metadata that was written at the time of recording and make simultaneous adjustments to camera raw media throughout your project.

For a detailed explanation of each of the RAW camera parameters supported by DaVinci Resolve, *see Chapter 7, "Camera Raw Settings."*

File





The File Inspector controls

Metadata

The File panel of the Inspector provides a consolidated way to view and edit a subsection of a clip's most commonly used media file metadata. It's easily accessible in the Inspector across the Media, Cut, Edit, and Fairlight pages.

The tab is composed of the following parts:

Clip Details: Presents data about the clip's data format (codec, resolution, frame rate, etc.).

Metadata: Presents a reduced set of common metadata fields for quick user entry.

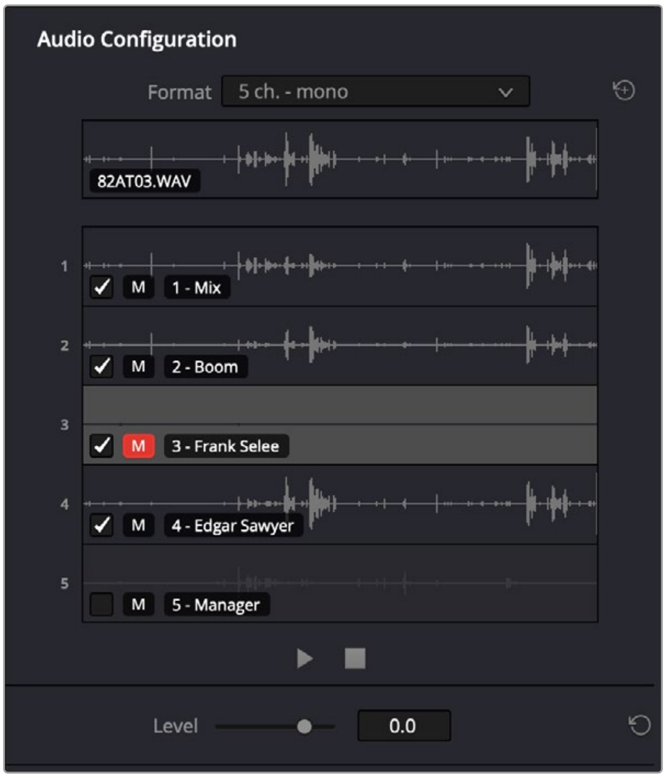
- **Timecode:** The start timecode of the clip. This field is editable if you want to manually change the clip's starting timecode.
- **Date Created:** The date that the clip was created. This field is editable if you want to manually change the clip's creation date.
- **Camera:** Sets the Camera # metadata.
- **Reel:** Sets the Reel/Card ID.
- **Scene:** The Scene number of the clip.
- **Shot:** The Shot letter/number of the clip.
- **Take:** The Take number of the clip.
- **Good Take:** This checkbox indicates if the clip is a good or circled take.
- **Clip Color:** Assign a specific color to a clip that is reflected in the Timeline.
- **Name:** This can be entered manually and changes a clip's name in that specific timeline only.
- **Comments:** Add a text description to the clip.
- **Auto Select Next Unsorted Clip:** When this box is checked, the next clip in the Media Pool is selected when you hit the Return button after entering a metadata field, and the cursor is automatically placed in the same field. This allows rapid sequential metadata entry without having to manually click to load each individual clip in the Media Pool. The Next Clip button will select the next clip in the Media Pool, regardless of the checkbox status.

Audio Configuration

The File tab in the Inspector now has an Audio Configuration pane that handles the controls that were formerly handled by Clip Attributes in the Media Pool (though that option is still available). The Audio Configuration pane provides a more intuitive and visual way of changing the track properties of an audio clip. Simply click on an audio clip in the Media Pool or Timeline, and then on the File Inspector to reveal this interface.

The pane features a per-channel waveform display for all tracks within a multichannel audio file. If the tracks have been named in the audio recorder, these names will appear over their respective tracks as well. The Audio Configuration panel can preview up to 36 tracks of audio.

At the top of the pane, a composite waveform is shown of all the tracks and is updated depending on the mute status of individual tracks. By default this composite is heard when the Play button is activated, and all channels are audible.



The Audio Configuration panel. Track 3 has been muted and Track 5 has been disabled.

A format menu allows you to choose common configurations for your source audio file without cumbersome manual re-routing. Custom routing can still be accommodated by choosing custom in the drop-down, which brings up the older clip attributes for situations that require less common configurations.

Audio for a selected source or timeline clip can be played or skimmed by moving the cursor along the waveform, and the specific track is solo'ed when skimming or playing. The play position or track being monitored can also be switched dynamically during playback. For example, you can start playback on track one, then simply click on track two, and the playback continues from that position. These controls let you quickly skim through and identify exactly what audio is on which track for further adjustment.

Each track has two adjustments that can be made, an Enable/Disable checkbox and a Mute button.

Enable/Disable: Enabling a track makes that track available for use in editing operations. Disabling a track removes that track from use in editing operations. For example, if you disable Track 2 of a 4-track audio file, when you drag that audio file from the Media Pool into the timeline, only 3 tracks (1, 3, and 4) will come over.

These adjustments can only be made on Media Pool clips; audio clips already in the timeline will have these options disabled.

Mute: Clicking this icon will mute the track so it is unheard but leave the actual track in place when used in editing operations and dragged into the timeline. Option clicking a mute button on a channel will allow you to solo by muting all other channels.

Underneath the track layout is a simple transport control comprised of Play and Stop buttons to start and stop audio playback.

Adjusting Trim Levels of Individual Channels in a Source File

The level of individual source clip channels in the audio inspector can be be adjusted via a trim control. This allows source clip audio in one channel to be brought up in level or brought down in level, as needed.

The trimmed level is internal to source clip in the Media Pool and will be inherited whenever that source clip configuration is used on a timeline, unless it is changed.

> **NOTE:** The trim level is totally separate from clip gain in the timeline. Be aware that trimmed levels can clip if you move them too high.

You can adjust the trim levels of individual audio channels in a multichannel clip.

Using the Trim slider

1. Select one or more channels in the Audio Configuration section of the File Inspector
2. Adjust the Level slider at the bottom to the desired audio level.

The waveform will adjust dynamically to the level value and will display a trimmed value in dB in the channel itself. If you have multiple lanes selected, all channels will be trimmed together. Relative levels are not stored, so if you adjust one, it will then force any others in the selection to match. When no channel lanes are selected, the Level control is grayed out.

Multiple Clip Selection and Adjustments

You can select multiple audio clips and adjust their properties in the Audio Configuration pane. For example, you can select a group of audio files and remove Track 2 from all of them at once. However, the following should be noted:

- In a multiple clip selection, only the last selected clip will appear in the track layout of the Audio Configuration pane. However the top composite waveform will be named “Multiple Clips” to let you know that more than one clip has been selected.
- Any adjustments, like muting or enabling/disabling a track will be applied to all the selected clips at once.

Timecode

The File tab in the Inspector now has a Timecode pane that handles the types of controls that were formerly handled by Clip Attributes in the Media Pool (though that option is still available). Here you can override the timecode details for a clip or clips in the Media Pool.

- **Current Frame:** Lets you assign a new time for the timecode at the currently viewed frame of the clip.
- **Slate:** In situations where source media comes from a shoot where a timecode slate was used during the shoot, then you can assign the slate timecode as a second timecode track that can be used for various operations, without changing the primary timecode of the clip, which may already be in use for program sync.

  To set appropriate Slate timecode, select a clip in the Media Pool with a visible timecode slate, and move the playhead to a frame where the timecode in the slate is clearly readable. Then, open the Timecode panel of the Clip Attributes window, and type the timecode value you see in the image into the Slate Timecode field.

- **Offset Source:** If an entire set of clips has timecode that’s merely offset, you can correct the timecode offset for as many selected clips as you like.